

Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

Tianxin Zhou

Spring 2026

ECEN 525 – Brain–Computer Interaction

Santa Clara University

Problem: EEG is non-stationary – and the geometry is what changes

EEG varies on **three geometric axes**:

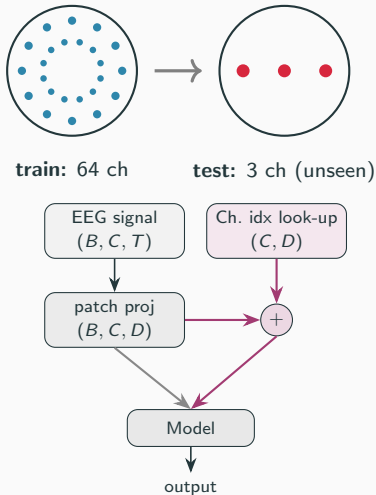
- **Session** — cap shifts day to day.
(*motivates*)
- **Subject** — head sizes differ.
(*motivates*)
- **Montage** — 2 to 256 electrodes in different layouts. ← **what I test**

Codex models (LaBraM, EEGPT):

identity-indexed embeddings — **no way to generalize** to an unseen layout.

Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



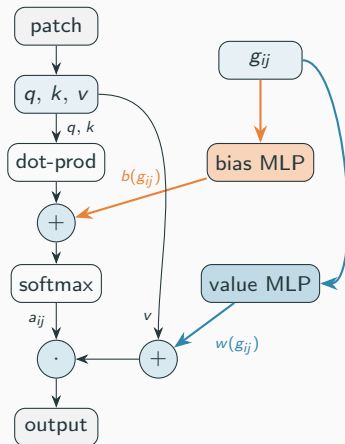
Solution: spatial attention conditioned on g_{ij}

$$g_{ij} = [p_i, p_j, p_i - p_j, \|p_i - p_j\|] \in \mathbb{R}^{10}$$

Example: $g_{C3 \rightarrow Cz}$ (normalized coords)

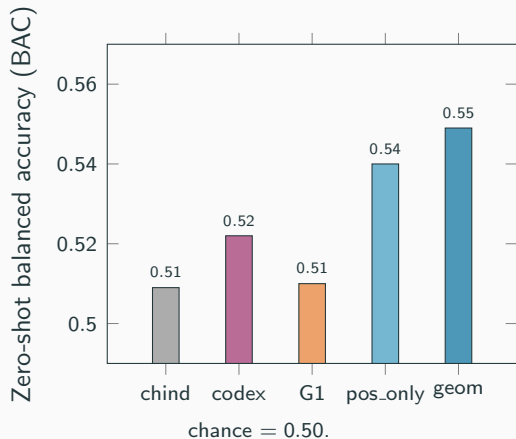
p_i (C3)	(x, y, z)	(-0.71, 0.00, 0.59)
p_j (Cz)	(x, y, z)	(0.00, 0.00, 0.95)
$p_i - p_j$	disp.	(-0.71, 0.00, -0.36)
$\ p_i - p_j\ $	dist.	0.80

$3 + 3 + 3 + 1 = 10$ dims. **pos_only** ablation
keeps just $[p_i, p_j] \in \mathbb{R}^6$.



bias path only = G1, **value path** only = G2, both = G3.

Results: geometric beats both baselines on zero-shot transfer



78 PhysioNet + 78 Sleep-EDFx pretrain → zero-shot to BCIC-2B

Paired across the 9 held-out BCIC-2B subjects

variant	BAC	<i>p</i>	95% CI of gain
chind	.509	—	—
G1	.510	.854	[−.008, +.010]
pos_only	.540	.022	[+.006, +.056]
geom	.549	.004	[+.017, +.063]